

PAUL SANSAN GYREYIRI

3D Computer Vision | Neural Rendering | Machine Learning |
Computer Vision Systems | XR/Immersive Computing | Applied AI Security

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Skills:

Technical Skills

- **3D Vision & Graphics**
3D Gaussian Splatting, Neural Rendering, NeRFs, Multi-view Geometry, Structure from Motion, PyTorch3D, COLMAP, Blender.
- **Computer Vision**
OpenCV, SAM2, Scene Text Detection, OCR Pipelines, Segmentation, Image Processing, Face Recognition.
- **Machine Learning & Deep Learning**
PyTorch, TensorFlow, Scikit-learn, CNNs, ResNet, RNNs, LSTMs, Transfer Learning, RFs, SVMs.
- **Programming**
Python, C++, Java, SQL, JavaScript, Kotlin.
- **XR & Software Systems**
Unity 3D, React.js, Node.js, Spring Boot, Git.
- **Data Pipeline Engineering:**
Large-scale dataset curation, preprocessing, quality control, annotation workflows.

Soft Skills

Collaboration, Leadership, Technical Communication, Research Documentation

Selected Projects:

RigidGS: Physics-Aware 3D Gaussian Splatting

- Designed an object-decoupling method for rigid-body interaction in 3D Gaussian Splatting.
- Integrated COLMAP, SAM2, and LaMa, and PyTorch3D into an end-to-end neural rendering pipeline.
- Designed custom boundary regularization and inpainting workflows for interaction-ready scene reconstruction.

LightLeaks — AR Optical Side-Channel Analysis

- Built computer vision pipelines for reconstructing sensitive content leaked through AR waveguide displays.
- Combined OCR, image enhancement, temporal fusion, and LLM reasoning for privacy inference.
- Evaluated reconstruction performance across multiple devices and capture conditions.

Experience:

University of Wyoming - Laramie – *Research Assistant*

January 2023 to Present

- Developed physics-aware neural rendering systems for interactive 3D scene reconstruction in VR environments. (ECCV 2026 under review).
- Built end-to-end 3D vision pipelines using COLMAP, PyTorch3D, SAM2 segmentation, LaMa inpainting, and neural rendering techniques for multi-object scene understanding.
- Designed multimodal computer vision pipelines integrating OCR, scene text detection, and LLM reasoning for AR privacy analysis across thousands of extracted ROI frames (ACM CCS under review).
- Engineered multimodal ML pipelines for VR/AR biometrics using gaze, EEG, and motion signals from 40+ participants across longitudinal sessions.
- Trained CNN and ResNet-based models for biometric authentication, achieving 2.9% Equal Error Rate on longitudinal EEG datasets.
- Published research in IEEE FG, ACM VRST, IEEE WIFS, with ongoing submissions to ECCV, ACM CCS, IJCB, and ACM TOPS.

University of Wyoming - Laramie– *Teaching Assistant*

January 2022 – January 2023, August 2023- May 2024

Facilitate laboratory sections. Schedule and maintain regular office hours to meet with students. Notify professor of errors or problems with assignments. Grade assignments and Labs. Courses handled includes Spring 2024: COSC 4555/COSC 5555, Machine Learning. Fall 2023: COSC 1030, Computer Science I (C++). Fall 2022: COSC 1030, Computer Science I (C++), and Spring 2022: COSC 1010, Introduction to Computer Science (Python).

Super Tech Limited - Accra – *Software Developer*

October 2021 to January 2022

Collected requirements from customers. Prepared and designed requirement documents. Undertake Development, Integration and Implementation of systems. Built and deployed microservices with spring boot. Built frontend applications using thyme leaf.

IPMC – Tarkwa – *Instructor*

March 2021 to September 2021

Taught Software Engineering Modules. Database Technology Modules and managed a consistent learning-focused classroom environment.

University of Mines and Technology -Tarkwa – *Teaching Assistant*

September 2020 to August 2021

Worked with Professors to monitor class schedule, supervise classroom activities. provided extra help to students., set up necessary materials for lectures and guided students through laboratory works.

Education:

Doctor of Philosophy in Computer Science | University of Wyoming

Ongoing (Expected July 2026)

Bachelor of Science in Computer Science and Engineering | University of Mines and Technology (Tarkwa, Ghana)

August 2020 | C.W.A: 81.08 (First Class Honors)

Relevant Coursework:

- **Computer Vision:** Advanced Image Processing, Pattern Recognition, 3D Computer Vision, Introduction to VR/AR
- **Machine Learning & AI:** Machine Learning, Artificial Intelligence, Blockchain Design
- **Systems:** Analysis of Algorithms, Distributed Algorithms, Cybersecurity, Computer Architecture
- **Mathematics:** Linear Algebra, Probability and Statistics, Discrete Mathematics, Mathematical Analysis

Selected Research and Publications:

- **RigidGS: Geometry-Aware Object Decoupling for Rigid-Body Interaction in 3D Gaussian Splatting**
Under review at ECCV 2026 [3D reconstruction, neural rendering, physics simulation, VR interaction]
- **LightLeaks: Sensitive Web Information Exposure through Light Leakage in Mixed Reality Displays**
Under review at ACM CCS 2026 [Computer vision, scene text detection, multimodal reasoning, AR privacy]
- **Domain-Informed Eye Movement Biometrics with Practical Evaluation for VR/AR User Authentication**
IEEE WIFS 2025
- **Head Movement Biometrics for Continuous Authentication in Virtual Reality**
ACM VRST 2025
- **HM-Auth: Redefining User Authentication in Immersive Virtual World Through Hand Movement Signatures**
IEEE FG 2024
- **Beyond Domain Adaptation: Understanding Feature Temporal Instability in EEG Authentication**
Under review at IJCB 2026
- **VR-Gate: Behavior as a Key to Seamless User Authentication in Virtual Reality**
Under review at ACM TOPS. [Traditional ML, Behavioral biometrics, Data Processing, VR security]